

**A Project Report On
UPI FRAUD DETECTION**

USING ML AND

EXPLAINABLE AI

*Submitted in partial fulfillment of
the requirement for the award of the
degree of*

BACHELOR OF COMPUTER APPLICATION



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CANDIDATE'S DECLARATION

We hereby certify that the work which is being presented in the project, entitled “**UPI Fraud Detection using ML and Explainable AI**” in partial fulfillment of the requirements for the award of the BCA (Bachelor of Computer Application) submitted in the School of Computer Applications and Technology of Galgotias University, Greater Noida, is an original work carried out during the period of Febuary, 2026 to June 2026, under the supervision of Dr. Somenath Bera Department of School of Computer Applications and Technology , Galgotias University, Greater Noida.

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Mr. Somenath Bera

CERTIFICATE

This is to certify that Project Report entitled “ **UPI Fraud Detection using ML and Explainable AI.**” which is submitted by **Sumit Kumar Karn, Aman Rathore and Muskan Jaiswal** in partial fulfillment of the requirement for the award of degree BCA in Department of School of Computer Applications and Technology, Galgotias University, Greater Noida, India is a record of the candidate own work carried out by him/them under my supervision. The matter embodied in this thesis is original and has not been submitted for the award of any other degree.

Signature of Examiner(s)

Signature of Supervisor

Date: June, 2026
Place: Greater Noida

ABSTRACT

Unified Payments Interface (UPI) has become one of the most widely used digital payment systems in India. The increasing popularity of online transactions has also led to a significant rise in fraudulent activities such as phishing attacks, fake collect requests, OTP scams, account takeover attempts, and social engineering frauds. Traditional fraud detection systems often rely on predefined rules and are unable to adapt quickly to new fraud patterns.

The objective of this project is to develop an intelligent fraud detection system named UPI Fraud Shield ML that can identify potentially fraudulent UPI transactions using machine learning techniques. The system analyzes transaction information, authentication behavior, transaction history, geography-based signals, user behavioral patterns, and UPI handle intelligence indicators to estimate the probability of fraud.

The project uses a dataset containing transaction attributes such as amount, transaction type, authentication attempts, OTP behavior, session activity, account age, transaction history, and handle-related features. Data preprocessing techniques such as missing value handling, duplicate removal, label encoding, feature selection, and train-test splitting were performed before model training.

Multiple machine learning algorithms including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and Neural Networks were trained and evaluated. The trained models were integrated into an interactive Streamlit dashboard that allows users to enter transaction details and receive fraud probability predictions in real time. Additional modules such as prediction history, model analytics, ROC curve visualization, feature importance analysis, and SHAP-based explainability were also implemented.

Experimental results demonstrate that ensemble learning methods such as Random Forest and Gradient Boosting achieve excellent performance on the dataset. The developed system successfully predicts fraudulent transactions while also providing interpretable explanations for its decisions. The project highlights the potential of machine learning techniques in improving the security of digital payment systems and reducing financial fraud.

TABLE OF CONTENTS

DECLARATION.....	i
CERTIFICATE.....	ii
ACKNOWLEDGEMENT	iii
LIST OF FIGURES.....	iv
 CHAPTER 1 INTRODUCTION.....	 13
1.1. Problem Introduction	
1.1.1. Motivation	
1.1.2. Project Objectives	
1.1.3. Scope of the Project	
1.2. Related Previous Work	
1.3. Organization of the Report	
 CHAPTER 2 SOFTWARE REQUIREMENT SPECIFICATION.....	 21
2.1 Product Perspective	
2.1.1 System Interfaces	
2.1.2 User Interfaces	
2.1.3 Hardware Interfaces	
2.1.4 Software Interfaces	
2.1.5 Communication Interfaces	
2.1.6 Memory Constraints	
2.1.7 Operations	
2.1.8 Site Adaptation Requirements	
2.2 Product Functions	
2.3 User Characteristics	

- 2.4 Constraints
- 2.5 Assumptions and Dependencies
- 2.6 Apportioning of Requirements
- 2.7 Use Case Analysis

CHAPTER 3 SYSTEM DESIGN..... 30

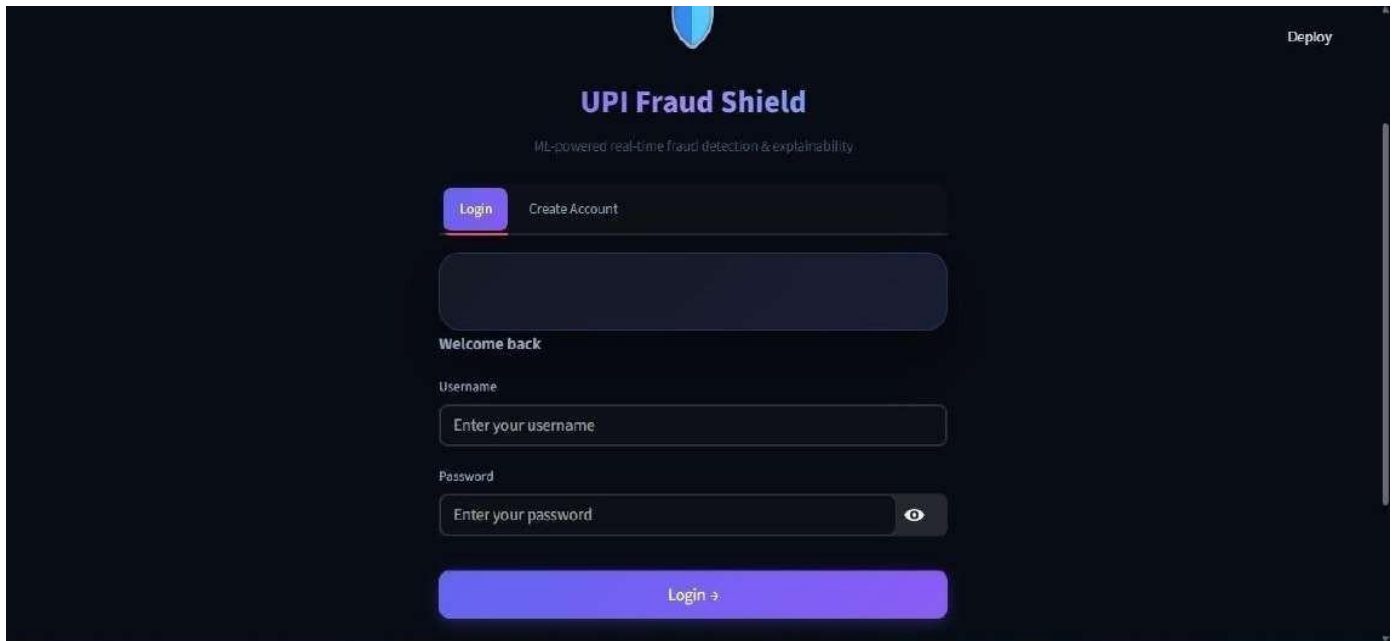
- 3.1 Architecture Diagram
- 3.2 Use Case Diagram
- 3.3 DFD Level 0
- 3.4 DFD Level 1
- 3.5 Activity Diagram
- 3.6 Sequence Diagram
- 3.7 Class Diagram
- 3.8 ER Diagram

CHAPTER 4 IMPLEMENTATION AND RESULTS 38

- 4.1 Software and Hardware Requirements
- 4.2 Assumptions and Dependencies
- 4.3 Constraints
- 4.4 Implementation Details
 - 4.4.1 Snapshots of Interfaces
 - 4.4.2 Test Cases
 - 4.4.3 Results

CHAPTER 5 CONCLUSION AND FUTURE ENHANCEMENT	55
5.1 Performance Evaluation	
5.2 Comparison with State-of-the-art Tech	
5.3 Conclusion	
5.4 Future Direction and Enhancements	

LIST OF FIGURE



The image shows a login page for 'UPI Fraud Shield'. At the top center is a blue shield icon. In the top right corner, there is a 'Deploy' link. The main heading is 'UPI Fraud Shield' in a bold, dark font, with the subtitle 'ML-powered real-time fraud detection & explainability' below it. There are two buttons at the top: 'Login' (highlighted with a red border) and 'Create Account'. Below these is a large, empty rectangular input field. Underneath the input field is the text 'Welcome back'. The 'Username' section has a label and an input field with the placeholder text 'Enter your username'. The 'Password' section has a label, an input field with the placeholder text 'Enter your password', and a toggle icon (an eye) to the right. At the bottom is a large, solid blue button with the text 'Login →'.

UPI Fraud Shield

ML-powered real-time fraud detection & explainability

[Login](#)[Create Account](#)

Create your account

Username

e.g. rahul_123

Email

you@email.com

Password

Min 6 characters

Confirm Password

Re-enter password

Create Account ->

**UPI Fraud Shield**
ML - Real-time - SHOP

[Predict Transaction](#)[My History](#)[Analytics](#)[Model Performance](#)

sumit_2004

sumitkarn2005@gmail.com

Logout

TRANSACTION DETAILS

TRANSACTION

Amount (₹)

12000.00

Transaction Type

pa... ▾

Merchant Category Code

5411

Hour of Day (0-23)

14

Session Source

app ▾

Fill in the transaction details and click
Analyze Transaction



CHAPTER 1- INTRODUCTION

1.1 Problem Introduction

The rapid advancement of information technology and the widespread adoption of digital payment systems have transformed the financial landscape across the world. In India, the introduction of the Unified Payments Interface (UPI) by the National Payments Corporation of India (NPCI) has revolutionized digital transactions by providing a fast, secure, and convenient platform for transferring money. UPI enables users to perform instant bank-to-bank transactions using mobile applications without the need to enter lengthy bank account details.

Over the past few years, UPI has emerged as one of the most widely used digital payment systems in India. Millions of transactions are processed daily through various applications such as Google Pay, PhonePe, Paytm, BHIM, and other banking applications. The ease of use, accessibility, and real-time processing capabilities of UPI have contributed significantly to its popularity among individuals, businesses, and organizations.

However, the rapid growth of digital payment systems has also resulted in an increase in cybercrime and financial fraud. Fraudsters continuously develop new methods to exploit users through phishing attacks, fake collect requests, QR code scams, social engineering techniques, account takeover attacks, malicious links, and OTP frauds. These fraudulent activities often lead to substantial financial losses and reduced trust in digital payment systems.

Traditional fraud detection systems mainly rely on predefined rules and manual verification processes. While these systems are effective in identifying known fraud patterns, they often fail to detect emerging and sophisticated fraud techniques. Furthermore, rule-based systems generate a large number of false positives, which can negatively impact user experience and operational efficiency.

Machine Learning has emerged as a powerful solution for fraud detection due to its ability to analyze large volumes of transaction data and identify hidden patterns. Machine learning algorithms can learn from historical transaction records and classify transactions as legitimate or fraudulent based on their characteristics. Such systems are capable of adapting to evolving fraud patterns and provide better prediction accuracy than traditional approaches.

This project, titled **UPI Fraud Shield ML: Intelligent UPI Fraud Detection System Using Machine Learning**, aims to develop an intelligent fraud detection system capable of identifying potentially fraudulent transactions using machine learning algorithms. The system analyzes multiple transaction- related, authentication-related, behavioral, and historical features to generate fraud probability predictions and provide explainable results.

The developed system incorporates multiple machine learning algorithms, performance evaluation metrics, analytics dashboards, prediction history management, and SHAP-based explainable artificial intelligence. The project demonstrates the practical application of machine learning techniques in enhancing the security of digital payment systems.

1.1.1 Motivation

The increasing number of digital payment frauds has become a significant concern for both users and financial institutions. Every year, thousands of individuals lose money due to fraudulent activities conducted through online payment platforms. As digital transactions continue to increase, the risk associated with cyber fraud also grows substantially.

Several motivating factors contributed to the development of this project.

The first motivation is the increasing prevalence of fraud in digital payment systems. Cybercriminals employ advanced techniques to deceive users and bypass traditional security mechanisms. Existing fraud detection systems often struggle to keep pace with evolving fraud strategies.

The second motivation is the need for intelligent decision-making systems. Machine learning algorithms have demonstrated remarkable success in various domains, including healthcare, cybersecurity, finance, and e-commerce. Applying machine learning to fraud detection can significantly improve the identification of suspicious transactions.

Another important motivation is the need for explainable fraud detection systems. Many machine learning models function as black boxes, making it difficult for users to understand why a particular transaction was classified as fraudulent. By integrating SHAP explainability techniques, the developed system provides transparency and improves user trust.

The project also serves as an opportunity to apply concepts learned in machine learning, data analytics, software engineering, and web application development. It combines theoretical knowledge with practical implementation and demonstrates the integration of multiple technologies into a real-world application.

Furthermore, the project contributes to enhancing awareness regarding financial fraud and demonstrates how modern technologies can be utilized to improve the security of digital transactions.

1.1.2 Project Objectives

The primary objective of this project is to develop an intelligent machine learning-based system capable of identifying fraudulent UPI transactions with high accuracy and reliability.

The specific objectives of the project are as follows:

1. To develop a fraud detection system capable of classifying transactions as legitimate or fraudulent.
2. To analyze transaction patterns and identify suspicious activities using machine learning algorithms.
3. To compare the performance of multiple machine learning models including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and Neural Network.
4. To evaluate model performance using metrics such as Accuracy, Precision, Recall, F1 Score, and ROC-AUC.
5. To develop an interactive web application using Streamlit for fraud prediction and analysis.
6. To provide fraud probability scores for better decision-making.
7. To implement prediction history functionality for maintaining previous transaction records.
8. To integrate SHAP-based explainability for interpreting model predictions.
9. To provide visual analytics through ROC curves, feature importance analysis, and model comparison dashboards.
10. To demonstrate the practical application of machine learning in financial fraud detection.

1.1.3 Scope of the Project

The scope of the project primarily focuses on the development and evaluation of a machine learning-based fraud detection system for digital payment transactions.

The system is designed to analyze transaction information and identify potentially fraudulent activities based on multiple behavioral and transactional features. The project covers the complete machine learning lifecycle, including data preprocessing, feature engineering, model training, evaluation, deployment, and explainability.

The developed application can be used as a prototype solution in various domains such as:

- **Banking Sector**

Banks process millions of digital transactions every day. Fraud detection systems can assist banks in identifying suspicious transactions and preventing financial losses.

- **Financial Technology Companies**

FinTech companies rely heavily on digital payment systems. Machine learning-based fraud detection solutions can enhance transaction security and improve customer trust.

- **Payment Gateways**

Payment gateway providers can integrate fraud detection models to monitor transactions and identify potentially malicious activities.

- **Educational and Research Applications**

The project serves as a valuable educational resource for understanding machine learning applications in cybersecurity and financial technology.

- **Digital Wallet Platforms**

Digital wallet providers can utilize fraud detection systems to improve the security of their payment ecosystems.

Although the current project operates in an offline environment using historical transaction data, the architecture can be extended to support real-time fraud monitoring and cloud-based deployment in future implementations.

1.2 Related Previous Work (Literature Survey)

Fraud detection has been an active area of research due to the increasing adoption of digital payment systems and the corresponding rise in cybercrime. Researchers and organizations have explored various approaches for identifying fraudulent transactions and minimizing financial losses.

Rule-Based Fraud Detection Systems

Early fraud detection systems primarily relied on rule-based mechanisms. These systems used predefined rules and thresholds to identify suspicious transactions. Examples include flagging transactions above a certain amount, detecting unusual login locations, or identifying multiple failed authentication attempts.

While rule-based systems are simple and easy to implement, they suffer from several limitations. They are unable to adapt to evolving fraud patterns and often generate a high number of false positives. As a result, researchers began exploring data-driven approaches for fraud detection.

Statistical Fraud Detection Techniques

Statistical techniques were introduced to improve fraud detection performance. These methods analyze transaction behavior and identify anomalies based on statistical deviations from normal patterns.

Statistical models provide greater flexibility than rule-based systems but often struggle with highly complex and dynamic datasets. Their effectiveness is limited when dealing with large-scale financial transaction data.

Machine Learning-Based Fraud Detection

Machine learning has become one of the most widely adopted approaches for fraud detection. Machine learning algorithms can automatically learn patterns from historical data and identify suspicious transactions without relying solely on manually defined rules.

1. Logistic Regression

Logistic Regression is one of the most commonly used classification algorithms. It estimates the probability that a transaction belongs to a fraudulent or legitimate class.

Advantages:

- Simple implementation
- Fast training
- High interpretability

Limitations:

- Limited ability to capture complex relationships
- Performance decreases with non-linear data.

2. Decision Tree

Decision Trees classify transactions through a sequence of decision rules. The algorithm recursively splits data based on feature values until classification is achieved.

Advantages:

- Easy visualization
- High interpretability
- Fast prediction

Limitations:

- Overfitting risk
- Sensitive to data variations

3. Random Forest

Random Forest is an ensemble learning technique that combines multiple decision trees to improve classification accuracy.

Advantages:

- High accuracy
- Reduced overfitting
- Robust performance

Limitations:

- Increased computational complexity
- Reduced interpretability compared to single trees

4. Gradient Boosting

Gradient Boosting builds multiple weak learners sequentially and improves prediction performance by correcting previous errors.

Advantages:

- Excellent predictive performance
- Effective handling of complex datasets

Limitations:

- Longer training time
- Sensitive to parameter selection

5. Neural Networks

Artificial Neural Networks are inspired by the structure of the human brain and are capable of learning highly complex relationships within data.

Advantages:

- Strong performance on complex datasets
- Ability to capture non-linear patterns

Limitations:

- High computational requirements
- Difficult to interpret

6. Explainable Artificial Intelligence

As machine learning models become increasingly complex, explainability has become an important area of research. Explainable Artificial Intelligence (XAI) techniques help users understand how models make decisions.

SHAP (SHapley Additive exPlanations) is one of the most popular explainability techniques.

SHAP calculates the contribution of each feature to the final prediction, thereby improving transparency and trust in machine learning systems.

The present project integrates SHAP explainability with machine learning-based fraud detection to provide both accurate predictions and meaningful explanations.

1.3 Organization of the Report

This report is organized into five chapters.

Chapter 1 introduces the project, discusses the problem statement, motivation, objectives, scope, and literature survey related to fraud detection systems.

Chapter 2 presents the Software Requirement Specification (SRS), including system requirements, interfaces, user characteristics, constraints, assumptions, and use case analysis.

Chapter 3 describes the system design, architecture, UML diagrams, data flow diagrams, activity diagrams, sequence diagrams, ER diagrams, and database schema.

Chapter 4 explains the implementation methodology, software and hardware requirements, machine learning models, test cases, user interface snapshots, and experimental results.

Chapter 5 presents the conclusion of the project, performance evaluation, comparison with existing technologies, and future enhancement opportunities.

The report concludes with references and appendices containing supporting information, screenshots, and additional project details.

CHAPTER 2-SOFTWARE REQUIREMENT SPECIFICATION

2.1 Product Perspective

The UPI Fraud Shield ML system is a machine learning-based web application developed to identify potentially fraudulent digital payment transactions. The system is designed to assist users, financial institutions, and researchers in analyzing transaction behavior and predicting fraud using advanced machine learning techniques.

Traditional fraud detection systems primarily depend on manually defined rules and threshold-based mechanisms. Such systems are often unable to detect newly emerging fraud patterns and may generate a large number of false alerts. The proposed system overcomes these limitations by utilizing machine learning algorithms capable of learning from historical transaction data and identifying complex fraud patterns.

The system operates through an interactive Streamlit-based web interface where users can enter transaction information and receive fraud predictions. Multiple machine learning models are used to classify transactions, while SHAP explainability techniques provide transparent explanations for model decisions. The system also maintains user authentication and prediction history for future analysis.

The proposed solution functions as a standalone application and can be further extended for integration with banking systems, payment gateways, and cloud-based fraud monitoring platforms.

2.1.1 System Interfaces

The UPI Fraud Shield ML system interacts with various internal modules to perform fraud prediction and analysis.

User Interface

The system provides a graphical user interface through which users interact with the application. Users can register, log in, enter transaction details, select machine learning models, and view fraud prediction results.

Machine Learning Interface

The machine learning interface processes transaction features and communicates with trained machine learning models to generate fraud predictions and probability scores.

Analytics Interface

The analytics interface generates performance visualizations such as ROC curves, feature importance graphs, model comparison charts, and prediction summaries.

Explainability Interface

The SHAP explainability module provides feature contribution analysis and explains why a transaction was classified as fraudulent or legitimate.

2.1.2 User Interfaces

The user interface has been designed to be simple, interactive, and user-friendly. The application consists of multiple screens that allow users to perform various operations.

Login Interface

The login interface allows registered users to securely access the system by entering valid credentials.

Registration Interface

The registration interface enables new users to create accounts by providing a username, email address, and password.

Fraud Prediction Dashboard

The fraud prediction dashboard allows users to enter transaction details and receive fraud predictions generated by machine learning models.

Analytics Dashboard

The analytics dashboard displays model performance metrics, ROC curves, feature importance graphs, and comparative analyses of machine learning algorithms.

History Dashboard

The history dashboard stores and displays previous fraud predictions for future reference.

The user interface is designed to provide clear navigation and intuitive interaction for users with varying levels of technical expertise.

2.1.3 Hardware Interfaces

The developed system does not directly communicate with specialized hardware devices. It operates on standard computing systems and requires only basic hardware resources. The hardware requirements include:

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB
- Storage: Minimum 500 MB free disk space
- Display: Standard monitor or laptop screen
- Input Devices: Keyboard and Mouse

Since the application is developed using Python and Streamlit, no dedicated hardware interfaces are required.

2.1.4 Software Interfaces

The application relies on several software components and libraries to perform its operations.

- **Python**
Python serves as the primary programming language used for developing the fraud detection system.
- **Streamlit**
Streamlit provides the web application framework used for developing the interactive user interface.
- **Scikit-Learn**
Scikit-Learn is utilized for machine learning model implementation, training, prediction, and evaluation.
- **Pandas**
Pandas is used for data manipulation, preprocessing, and feature management.
- **NumPy**
NumPy supports numerical computations and matrix operations.
- **SHAP**
SHAP provides explainable artificial intelligence capabilities and feature contribution analysis.
- **Plotly and Matplotlib**
These libraries are used for generating interactive visualizations and performance charts.
- **Joblib**
Joblib is used for saving and loading trained machine learning models

2.1.5 Communication Interfaces

The current implementation operates as a local web application and does not require external communication protocols.

The communication process occurs through:

- Local HTTP communication between the browser and Streamlit server.
- Internal communication between application modules.
- Data exchange between the user interface and machine learning models.

Future versions of the system may integrate REST APIs, banking services, and cloud communication protocols for real-time fraud monitoring.

2.1.6 Memory Constraints

The system has been designed to operate efficiently on standard personal computers. The minimum memory requirements are:

- RAM: 4 GB
- Available Storage: 500 MB
- Python Environment: 1 GB

The fraud prediction process requires loading machine learning models and processing transaction features. The memory consumption remains relatively low and is suitable for educational and research applications.

2.1.7 Operations

The system supports multiple operational functionalities.

- **User Registration**
Users can create accounts using valid credentials.
- **User Authentication**
Registered users can securely log into the application.
- **Fraud Prediction**
Users can enter transaction details and generate fraud predictions.
- **Probability Calculation**
The system computes fraud probability scores indicating the likelihood of fraudulent behavior.
- **Explainable AI Analysis**
The system generates SHAP explanations to justify prediction decisions.

- **Analytics Generation**
The system produces visual reports and performance metrics.
- **Prediction History Management**
Users can view, save, and manage previously generated predictions.
- **Logout Operation**
Users can securely terminate their active sessions.

2.1.8 Site Adaptation Requirements

Before deploying the system, the following setup requirements must be satisfied:

- **Python Installation**
Python version 3.10 or later must be installed.
- **Library Installation**
Required Python libraries must be installed using the requirements file.
- **Model Availability**
Trained machine learning model files must be available within the project directory.
- **Streamlit Configuration**
The Streamlit environment should be configured properly before execution.
- **User Data Storage**
The application requires access to JSON files used for storing user information and prediction history.

2.2 Product Functions

The UPI Fraud Shield ML system performs several important functions.

- **Authentication Function**
This function manages user registration, login, and account validation.
- **Fraud Detection Function**
This function analyzes transaction details and classifies transactions as fraudulent or legitimate.
- **Machine Learning Function**
Multiple machine learning models are used to improve prediction reliability and performance comparison.
- **Explainability Function**
The SHAP module explains the influence of individual features on fraud predictions.
- **Analytics Function**
The analytics module generates visual reports and performance comparison

- **History Function**

Prediction records are stored and displayed for future analysis.

2.3 User Characteristics

The system is intended for a diverse group of users.

- **Students**

Students can use the system to learn machine learning concepts and fraud detection techniques.

- **Researchers**

Researchers can analyze fraud detection methodologies and evaluate model performance.

- **Banking Professionals**

Banking professionals can study transaction behavior and fraud trends

- **Financial Analysts**

Financial analysts can utilize fraud prediction results for decision-making and risk assessment.

The application is designed to accommodate users with both technical and non-technical backgrounds.

2.4 Constraints

The project has several constraints.

- **Data Dependency**

The effectiveness of the fraud detection system depends heavily on the quality and diversity of the dataset.

- **Offline Environment**

The current system operates using historical transaction data and does not support real-time transaction monitoring.

- **Computational Resources**

Model training performance depends on available computational resources.

- **Fraud Evolution**

Fraud techniques continuously evolve, requiring periodic retraining of machine learning models.

- **Security Limitations**

The application serves as an educational and research prototype rather than a production-grade banking security system.

2.5 Assumptions and Dependencies

The development of the system is based on the following assumptions:

- Users provide valid transaction information.
- The dataset accurately represents real-world transaction patterns.
- Trained machine learning models are available.
- Required software libraries are installed.
- The execution environment remains stable.

Dependencies include:

- Python Runtime Environment
- Streamlit Framework
- Scikit-Learn Library
- SHAP Library
- Plotly Visualization Library
- Pandas and NumPy Libraries

2.6 Apportioning of Requirements

Some requirements have been reserved for future implementation.

- **Cloud Deployment**

The application may be deployed on cloud platforms such as AWS, Azure, or Google Cloud.

- **Mobile Application**

A mobile application version can be developed for Android and iOS devices.

- **Real-Time Fraud Monitoring**

Future versions may support real-time transaction monitoring through banking APIs.

- **Deep Learning Integration**

Advanced deep learning architectures can be incorporated for improved prediction accuracy.

- **Continuous Learning**

The system may be enhanced to continuously update models using new transaction data.

2.7 Use Case Analysis

The primary use cases of the system include:

- **User Registration**

Users create accounts and gain access to system functionality.

- **User Login**

Users authenticate themselves using valid credentials.

- **Fraud Prediction**

Users enter transaction details and receive fraud prediction results.

- **View Analytics**

Users analyze machine learning performance metrics and visual reports.

- **View Prediction History**

Users access previously generated fraud predictions.

- **Logout**

Users securely exit the application.

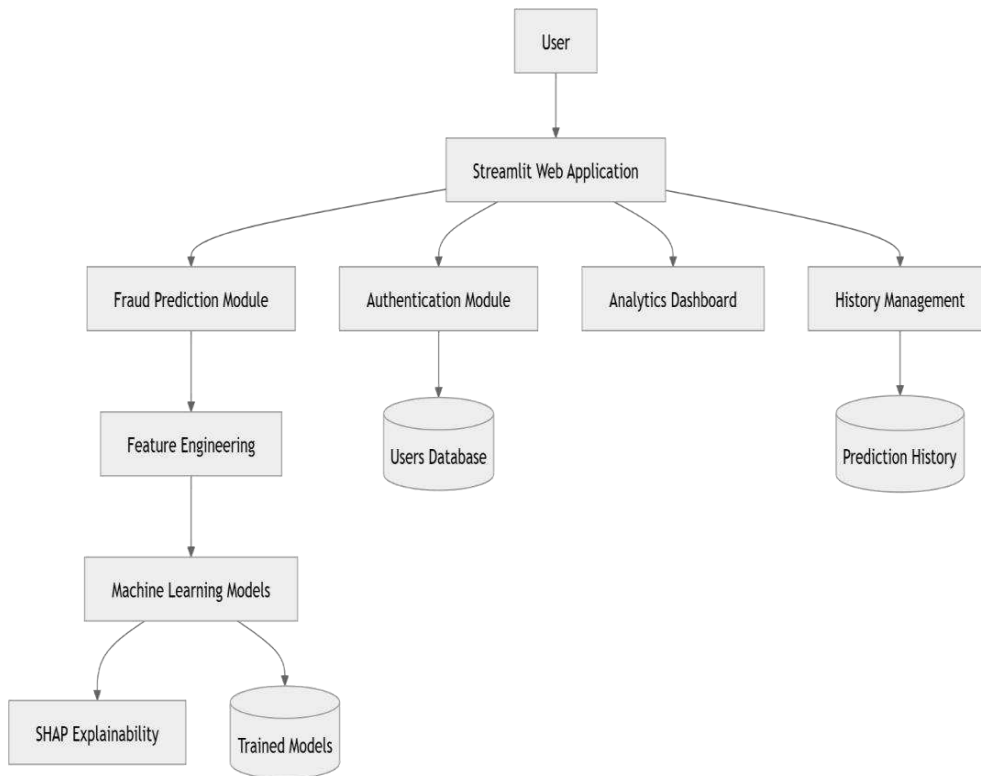
2.8 Sequence Diagram Description

The fraud prediction process follows a sequential workflow.

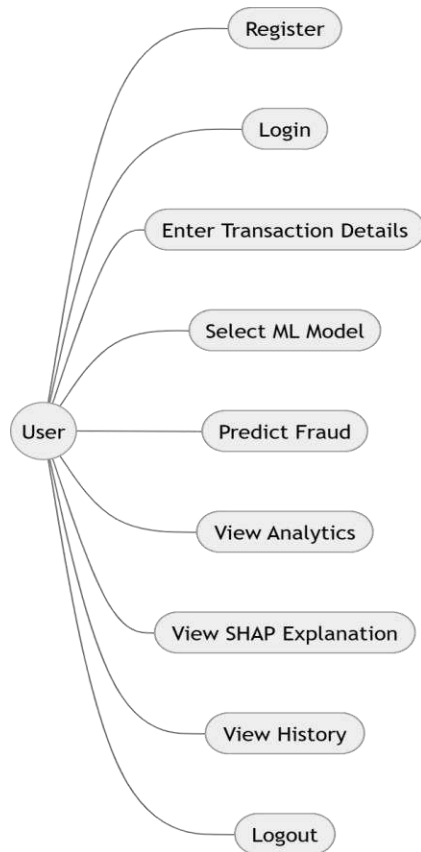
1. The user logs into the application.
2. Transaction information is entered.
3. Input data is validated.
4. Features are processed and transformed.
5. Selected machine learning models generate predictions.
6. Fraud probability scores are calculated.
7. SHAP explanations are generated.
8. Results are displayed to the user.
9. Prediction records are stored in history.
10. Users can review analytics and previous predictions.

Chapter 3: System Design

3.1 Architecture Diagram



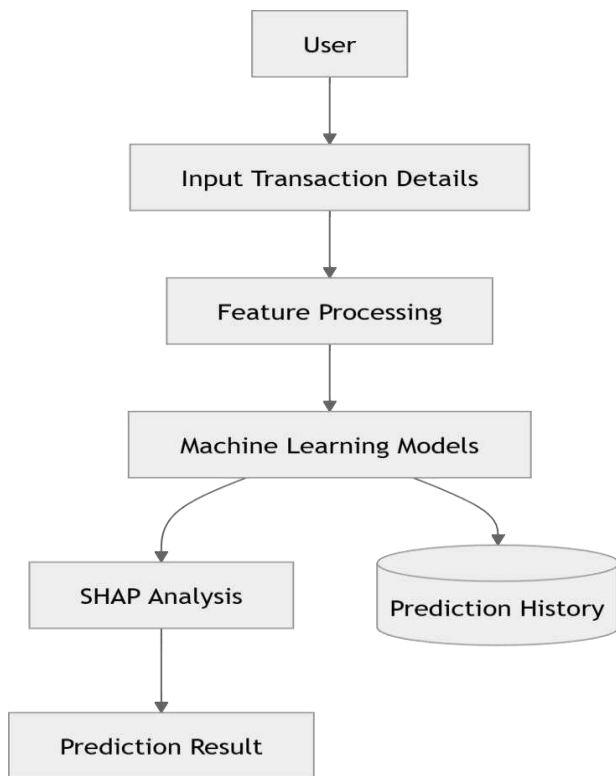
3.2 Use Case Diagram



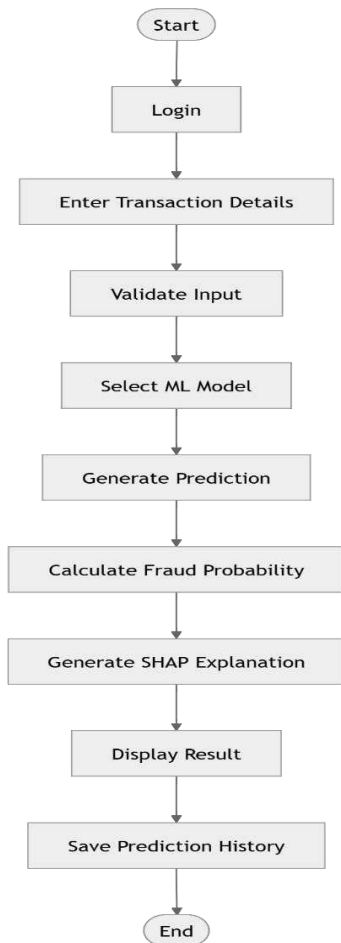
3.3 DFD Level 0



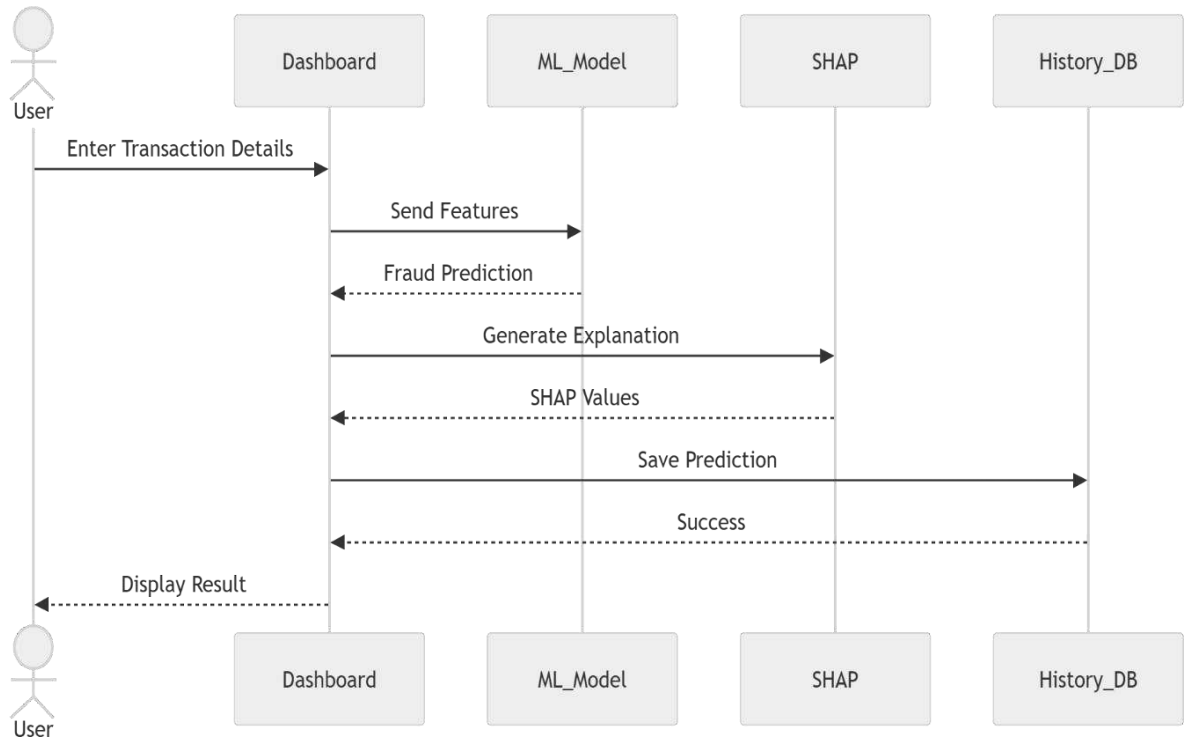
3.4 DFD Level 1



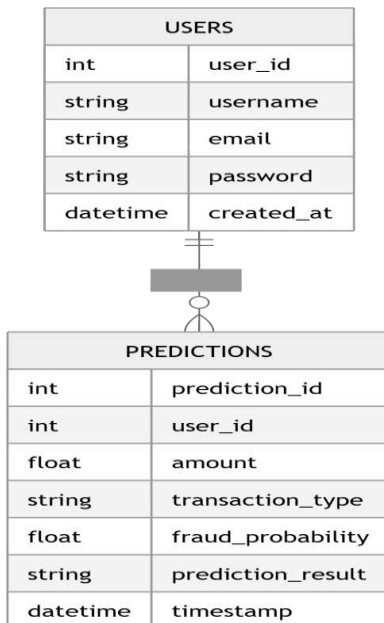
3.5 Activity Diagram



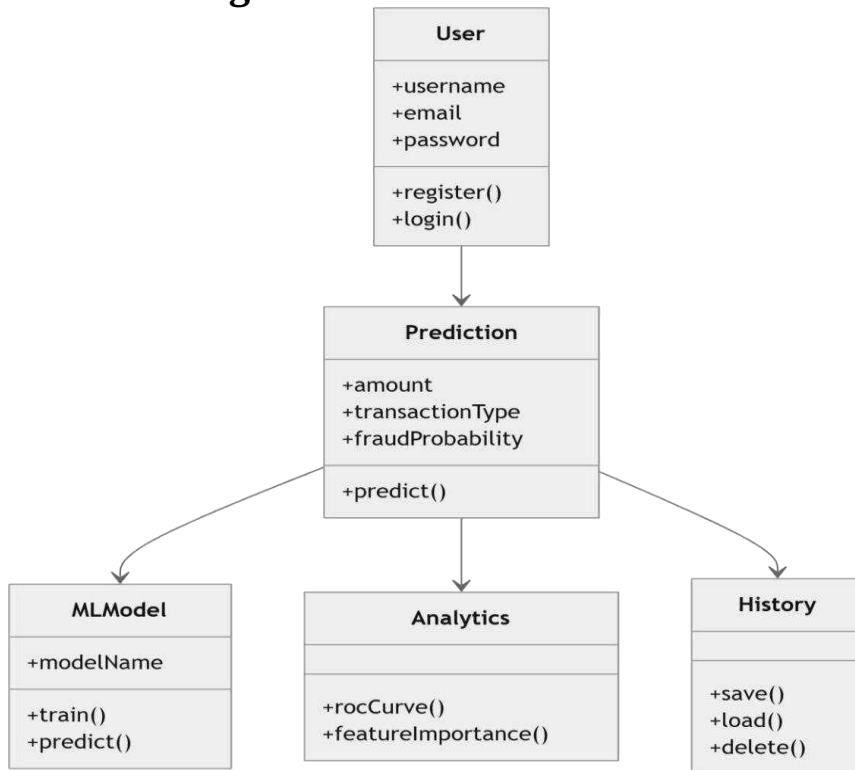
3.6 Sequence Diagram



3.7 Class Diagram



3.8 ER Diagram



CHAPTER 4- IMPLEMENTATION AND RESULTS

4.1 Software and Hardware Requirements

1. Software Requirements

The UPI Fraud Shield ML system was developed using modern machine learning and web application technologies. The software environment provides support for data preprocessing, model training, prediction generation, visualization, and explainable artificial intelligence.

The primary software requirements are:

Software	Purpose
Python 3.10+	Core programming language
Streamlit	Web application development
Pandas	Data preprocessing and manipulation
NumPy	Numerical computations
Scikit-Learn	Machine learning algorithms
SHAP	Explainable AI
Plotly	Interactive visualizations
Matplotlib	Graph plotting
Joblib	Model serialization

2. Hardware Requirements

The developed application can run efficiently on a standard personal computer.

Hardware Component	Specification
Processor	Intel Core i3 or higher
RAM	Minimum 4 GB
Storage	Minimum 500 MB free space
Display	1366 × 768 Resolution
Internet	Optional

The system does not require specialized hardware and can operate on most modern computers.

4.2 Assumptions and Dependencies

The project was developed based on several assumptions.

1. Users provide valid transaction information.
2. The dataset accurately represents real-world fraud scenarios.
3. Trained machine learning models are available during prediction.
4. Required Python libraries are installed successfully.
5. The execution environment remains stable. The

system depends on:

- Python Runtime Environment
- Streamlit Framework
- Machine Learning Libraries
- SHAP Library
- Dataset Availability
- Saved Model Files

These dependencies are necessary for proper functioning of the application.

4.3 Constraints

Several constraints were encountered during project development.

- **Dataset Constraints**

The quality of predictions depends heavily on dataset quality and diversity.

- **Computational Constraints**

Training complex machine learning models requires computational resources and processing time.

- **Security Constraints**

The current implementation is a prototype and is not directly integrated with banking systems.

- **Real-Time Constraints**

The application currently works with historical transaction data and does not process live banking transactions.

- **Scalability Constraints**

Future deployment may require cloud infrastructure to support large-scale transaction monitoring.

4.4 Implementation Details

The implementation phase involved data preprocessing, feature engineering, model training, evaluation, deployment, and visualization.

The complete system consists of several modules:

- **Authentication Module**

This module manages user registration and login functionality. User credentials are securely stored and validated before access is granted.

- **Fraud Prediction Module**

This module processes transaction information and generates fraud predictions using trained machine learning models.

- **Analytics Module**

The analytics module provides visual representations of model performance, feature importance, and fraud probabilities.

- **Explainability Module**

SHAP explainability techniques are integrated to explain model decisions and improve transparency.

- **History Management Module**

Prediction results are stored and can be reviewed by users through the history dashboard.

The modular architecture improves maintainability, scalability, and future enhancement possibilities.

4.4.1 Snapshots of Interfaces

1. Login Interface

The Login Interface allows registered users to securely access the system. Users

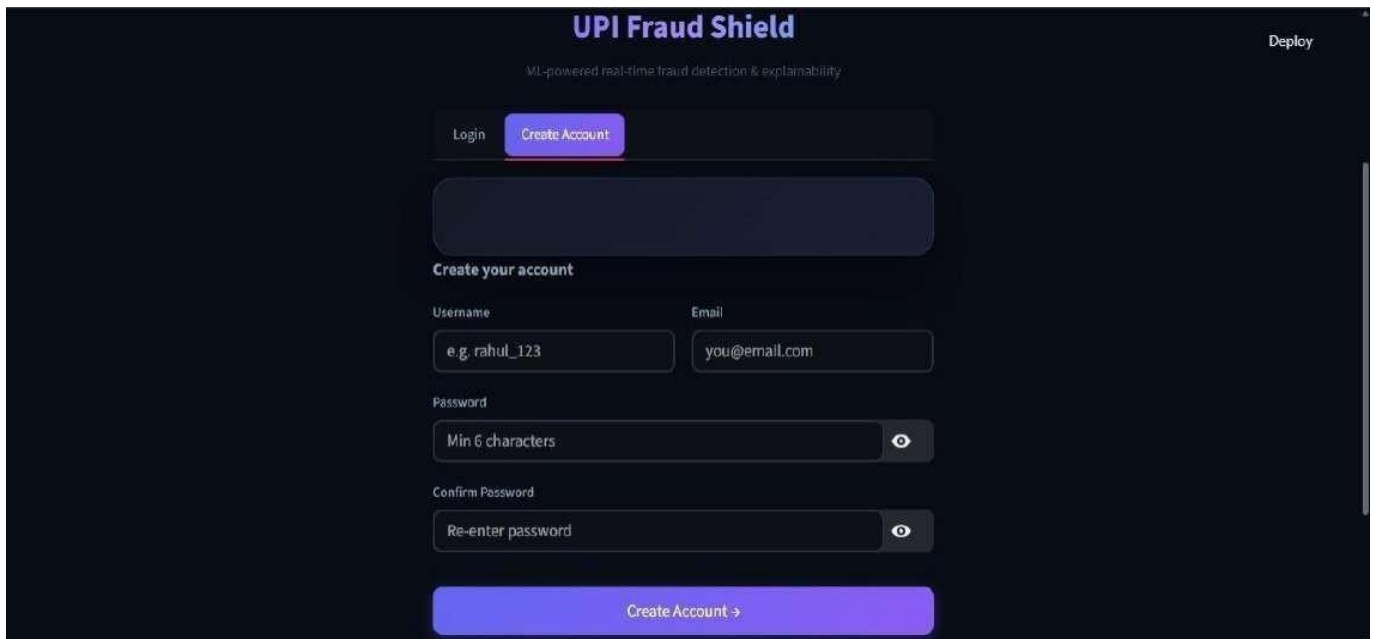
enter their username and password to authenticate themselves before using fraud prediction services.



The image shows the login interface of the 'UPI Fraud Shield' application. At the top, there is a blue shield icon and the title 'UPI Fraud Shield' in bold. Below the title, a subtitle reads 'ML-powered real-time fraud detection & explainability'. In the top right corner, there is a 'Deploy' button. The main content area features a 'Login' button (highlighted in red) and a 'Create Account' link. Below these, there is a large, empty, rounded rectangular input field. Underneath this field, the text 'Welcome back' is displayed. The 'Username' section has a label and an input field with the placeholder text 'Enter your username'. The 'Password' section has a label, an input field with the placeholder text 'Enter your password', and an eye icon to toggle visibility. At the bottom, there is a large red button labeled 'Login →'.

2. Registration Interface

The Registration Interface allows new users to create accounts. The system validates user details before creating an account.



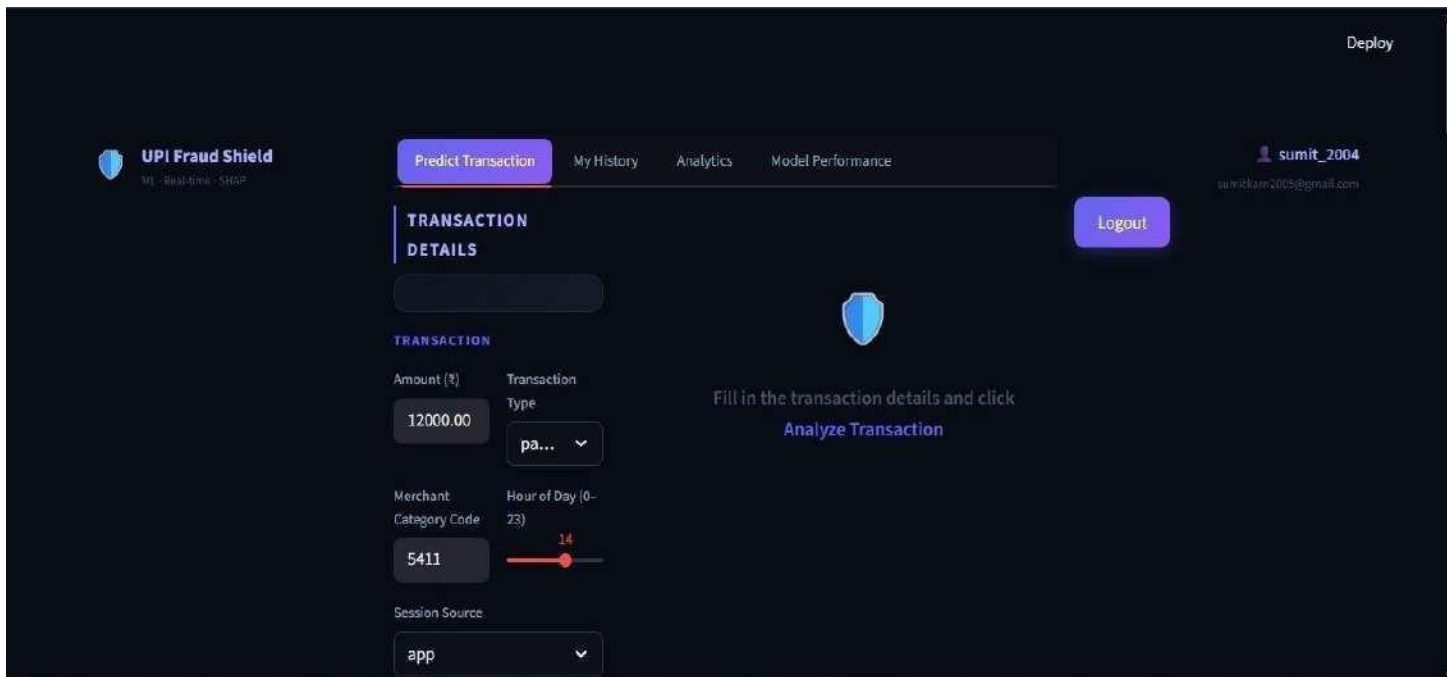
The image shows the registration interface of the 'UPI Fraud Shield' application. At the top, there is a blue shield icon and the title 'UPI Fraud Shield' in bold. Below the title, a subtitle reads 'ML-powered real-time fraud detection & explainability'. In the top right corner, there is a 'Deploy' button. The main content area features a 'Login' link and a 'Create Account' button (highlighted in red). Below these, there is a large, empty, rounded rectangular input field. Underneath this field, the text 'Create your account' is displayed. The 'Username' section has a label and an input field with the placeholder text 'e.g. rahul_123'. The 'Email' section has a label and an input field with the placeholder text 'you@email.com'. The 'Password' section has a label, an input field with the placeholder text 'Min 6 characters', and an eye icon to toggle visibility. The 'Confirm Password' section has a label and an input field with the placeholder text 'Re-enter password', and an eye icon to toggle visibility. At the bottom, there is a large red button labeled 'Create Account →'.

3. Fraud Prediction Dashboard

The Fraud Prediction Dashboard is the primary interface of the application. Users enter transaction details and obtain fraud predictions.

Features include:

- Transaction Input
- Fraud Probability
- Fraud Classification
- Model Selection

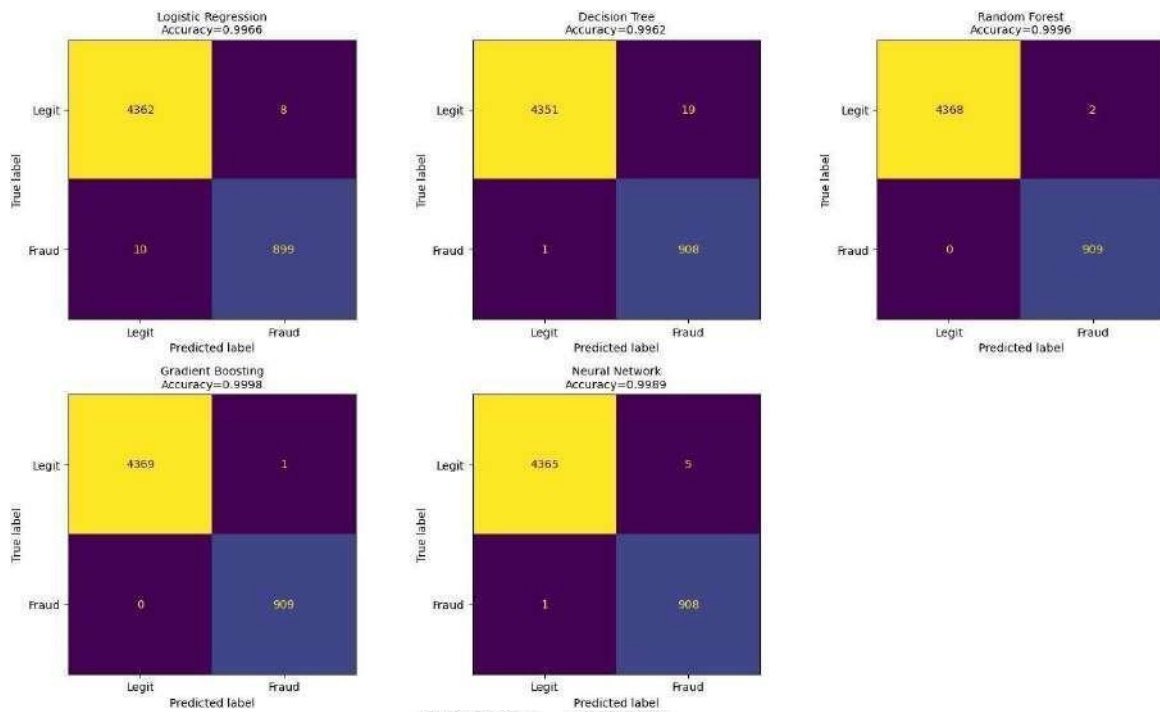


The screenshot displays the 'UPI Fraud Shield' dashboard. At the top left is the logo 'UPI Fraud Shield' with the tagline 'ML - Real-time - SHAP'. To the right are navigation tabs: 'Predict Transaction' (highlighted in red), 'My History', 'Analytics', and 'Model Performance'. In the top right corner, there is a 'Deploy' button and a user profile section for 'sumit_2004' with the email 'sumitkare2005@gmail.com' and a 'Logout' button. The main section is titled 'TRANSACTION DETAILS' and contains several input fields: 'Amount (%)' with the value '12000.00', 'Transaction Type' with a dropdown menu showing 'pa...', 'Merchant Category Code' with the value '5411', 'Hour of Day (0-23)' with a slider set to '14', and 'Session Source' with a dropdown menu showing 'app'. A large blue shield icon is centered in the background. Below the input fields, there is a text prompt 'Fill in the transaction details and click' followed by a red 'Analyze Transaction' button.

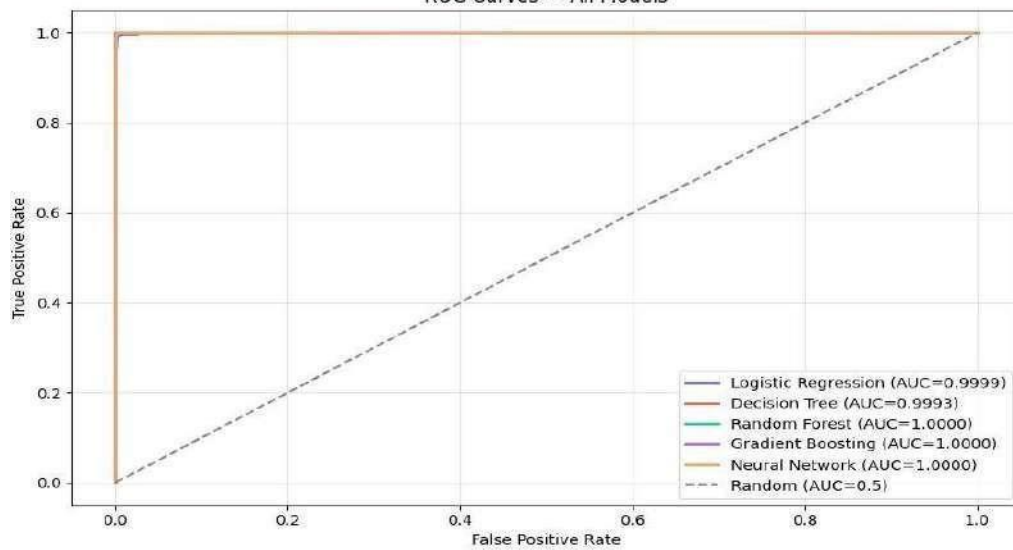
4. Analytics Dashboard

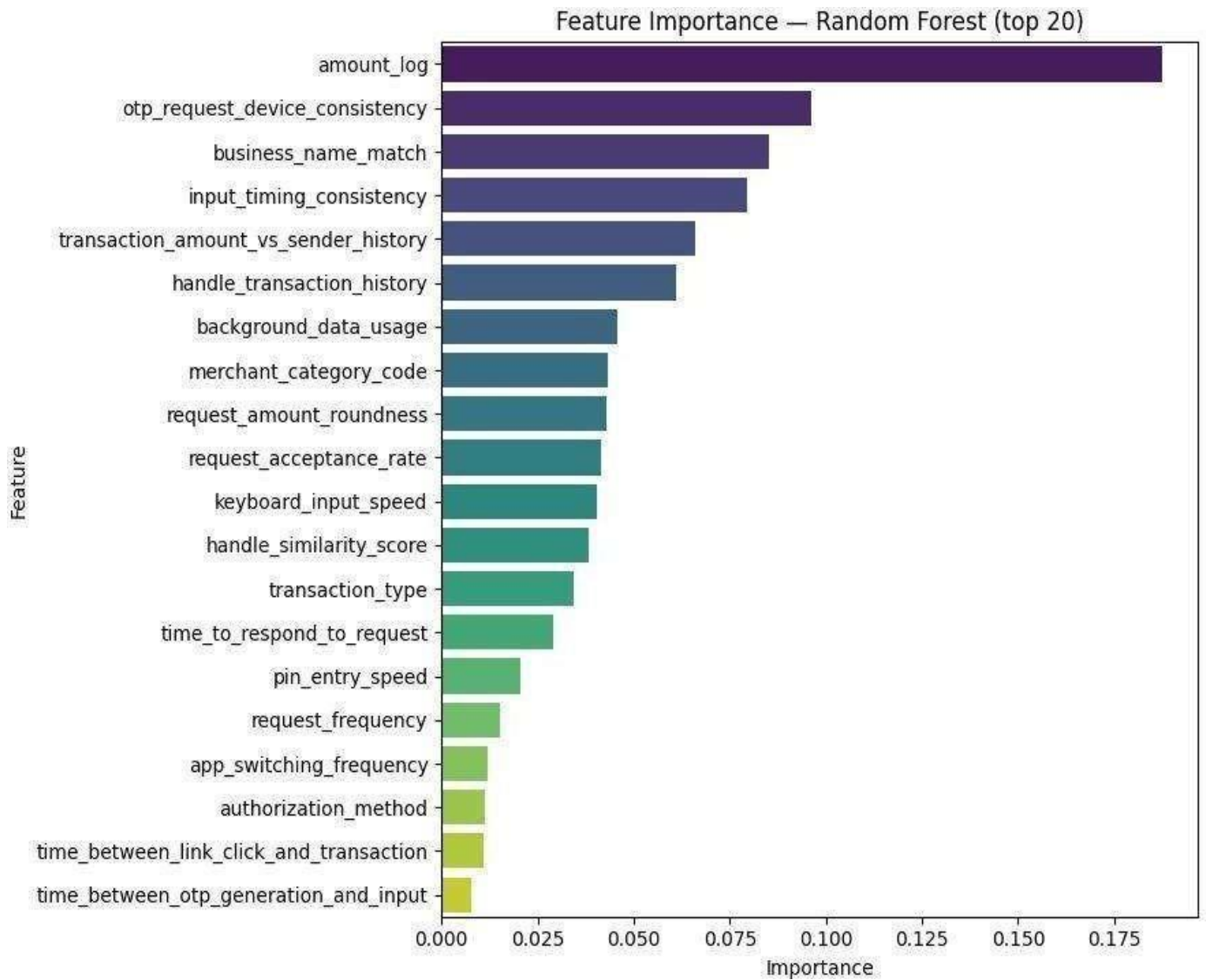
The Analytics Dashboard displays:

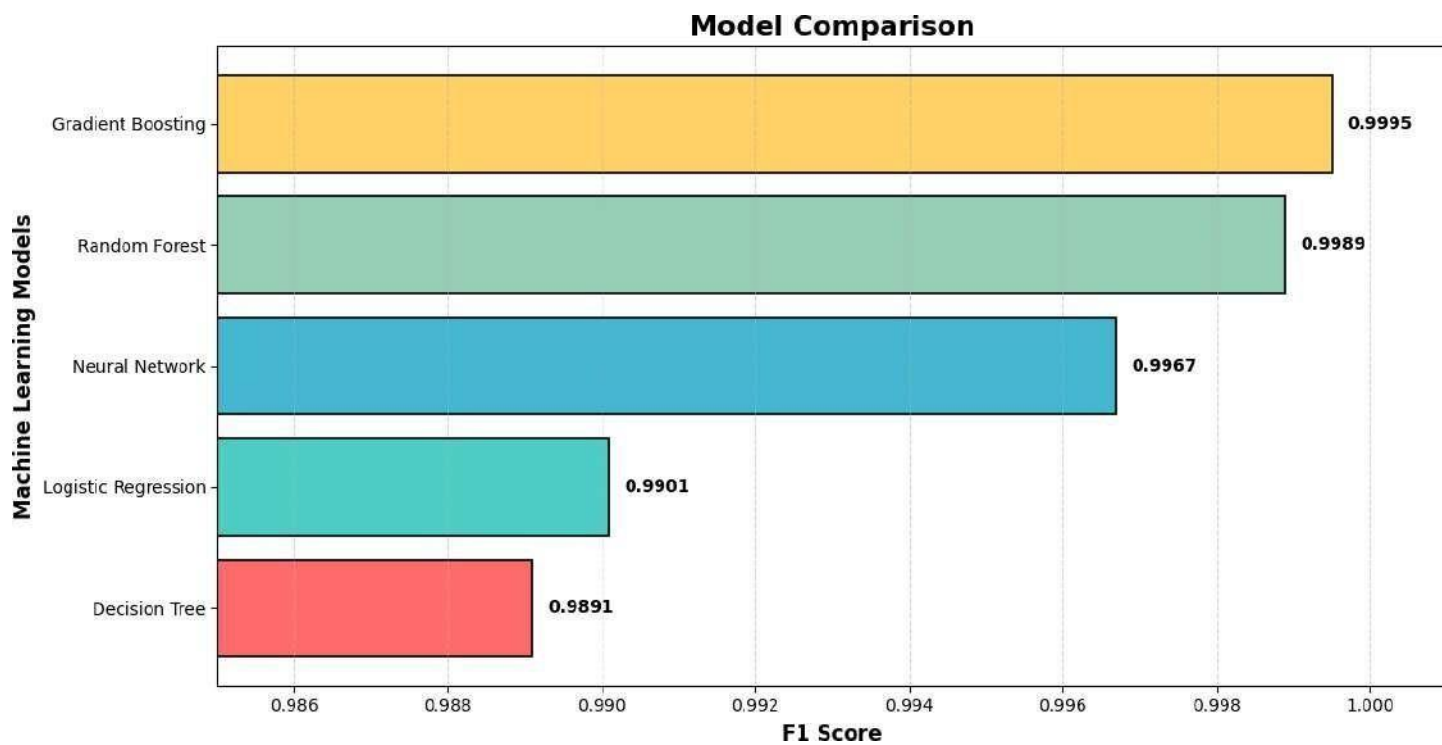
- ROC Curve
- Model Comparison
- Feature Importance
- Performance Metrics



ROC Curves — All Models







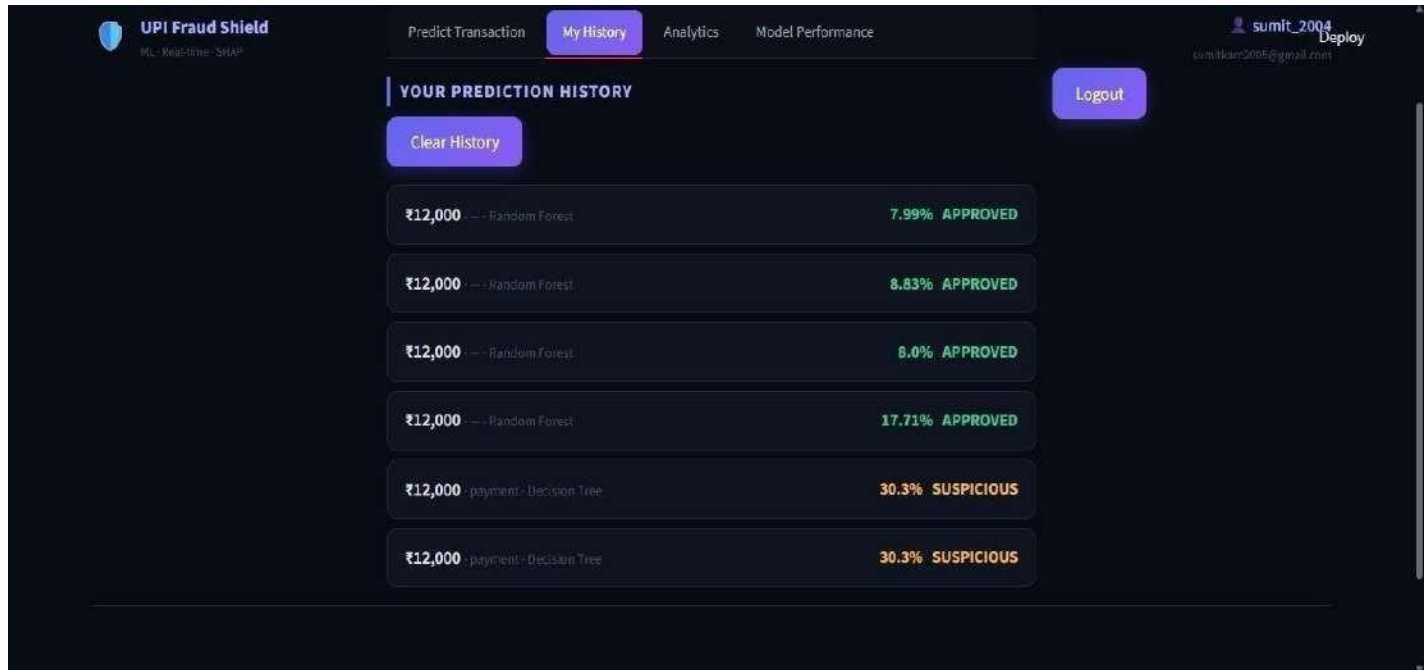
5. SHAP Explainability Dashboard

The SHAP dashboard provides feature contribution analysis and explains the reasons behind fraud predictions.



6. Prediction History Dashboard

The History Dashboard stores previous prediction results and allows users to review earlier transactions.



4.4.2 Test Cases

Several test cases were performed to verify system functionality.

Test Case 1: User Registration Test

Case ID	TC01
Objective	Verify user registration
Input	Valid user details
Expected Result	Account created successfully
Actual Result	Passed

Test Case 2: User Login Test

Case ID	TC02
Objective	Verify login functionality

Test Case ID**TC02**

Input Valid credentials Expected
Result Successful login Actual Result
Passed

Test Case 3: Fraud Prediction Test**Case ID****TC03**

Objective Generate fraud prediction
Input Transaction details
Expected Result Fraud classification Actual
Result Passed

Test Case 4: SHAP Explanation Test**Case ID****TC04**

Objective Generate SHAP explanation
Input Predicted transaction Expected
Result Feature contribution displayed Actual
Result Passed

Test Case 5: Analytics Dashboard Test**Case ID****TC05**

Objective Display analytics
Input Trained model results
Expected Result Charts generated Actual
Result Passed

Test Case 6: Prediction History

Test Case ID	TC06
Objective	Store prediction history
Input	Fraud prediction result
Expected Result	Record stored successfully
Actual Result	Passed

4.4.3 Results

The developed system was evaluated using multiple machine learning algorithms. The following models were implemented:

1. Logistic Regression
2. Decision Tree
3. Random Forest
4. Gradient Boosting
5. Neural Network

Performance Metrics

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC Score



1. Model Comparison

The comparison of different machine learning models demonstrated strong predictive performance.

Model	Accuracy	Precision	Recall	F1 Score	AUC - ROC
Logistic Regression	0.9966	0.9912	0.9890	0.9901	0.9999
Decision Tree	0.9962	0.9795	0.9989	0.9891	0.9993
Random Forest	0.9996	0.9978	1.0000	0.9989	1.0000
Gradient Boosting	0.9998	0.9989	1.0000	0.9995	1.0000
Neural Network	0.9989	0.9945	0.9989	0.9967	1.0000

2. ROC Curve Analysis

Receiver Operating Characteristic (ROC) curves were generated to evaluate classifier performance.

A higher Area Under Curve (AUC) indicates better discrimination between fraudulent and legitimate transactions.

3. Feature Importance Analysis

Feature Importance analysis was performed using Random Forest. Important features included:

- Receiver Account Age
- Receiver Transaction History
- Transaction Amount
- Request Amount Roundness
- OTP Request Device Consistency
- Authentication Attempts

These features contributed significantly to fraud prediction accuracy.

4. SHAP Explainability Results

SHAP values were generated to explain model decisions.

The SHAP analysis demonstrated how individual transaction features influenced fraud predictions. This enhanced transparency and increased confidence in model outputs.

CHAPTER 5- CONCLUSION AND FUTURE ENHANCEMENT

5.1 Performance Evaluation

The primary objective of this project was to develop an intelligent machine learning-based fraud detection system capable of identifying fraudulent digital payment transactions accurately and efficiently. The developed UPI Fraud Shield ML system successfully achieved this objective by integrating machine learning algorithms, data analytics, explainable artificial intelligence, and interactive visualization techniques.

The system was evaluated using multiple machine learning algorithms including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and Neural Network. Various performance metrics such as Accuracy, Precision, Recall, F1 Score, and ROC-AUC Score were utilized to assess the effectiveness of these models.

The evaluation results demonstrated that machine learning algorithms are highly effective in detecting fraudulent transactions. The trained models successfully learned patterns from transaction data and distinguished between legitimate and fraudulent activities. The fraud prediction module provided reliable probability scores that assisted users in understanding transaction risks.

The integration of SHAP Explainable Artificial Intelligence significantly improved the interpretability of predictions. Users could understand how individual features influenced fraud predictions, thereby increasing transparency and trust in the system.

The analytics dashboard provided valuable insights into model performance, feature importance, fraud trends, and classification behavior. The visualization of performance metrics enabled easy comparison between machine learning models.

Overall, the developed system demonstrated strong predictive capabilities and effectively fulfilled the objectives defined during the project planning phase.

5.2 Comparison with Existing State-of-the-Art Technologies

Fraud detection systems have evolved significantly over the years. Traditional systems relied heavily on rule-based mechanisms and manual verification

processes. While such approaches were useful for detecting known fraud patterns, they often failed to identify sophisticated and emerging fraud techniques.

Existing Rule-Based Systems

Traditional fraud detection systems operate using predefined rules and threshold values. Transactions are flagged as suspicious when they violate certain conditions such as:

- High transaction amount
- Multiple failed login attempts
- Unusual transaction location
- Excessive transaction frequency

Although these systems are simple to implement, they suffer from several limitations.

Limitations of Existing Systems

- Inability to adapt to new fraud patterns
- High false positive rates
- Manual rule maintenance
- Limited scalability
- Reduced prediction accuracy

Machine Learning-Based Systems

Modern fraud detection systems increasingly rely on machine learning algorithms to identify suspicious behavior. Machine learning models learn from historical transaction data and automatically identify hidden fraud patterns.

The proposed UPI Fraud Shield ML system utilizes machine learning techniques to improve fraud detection accuracy and adaptability.

Comparison Table

Feature	Traditional System	Proposed System
Detection Method	Rule-Based	Machine Learning
Adaptability	Low	High
Prediction Accuracy	Moderate	High
Fraud Pattern Learning	No	Yes

Feature	Traditional System	Proposed System
Explainability	Limited	SHAP-Based
Analytics Dashboard	Not Available	Available
Prediction History	Limited	Available
Scalability	Moderate	High
Automation	Low	High

The comparison clearly demonstrates that the proposed system offers significant improvements over traditional fraud detection approaches.

By incorporating machine learning and explainable artificial intelligence, the developed solution provides enhanced fraud detection capabilities and supports better decision-making.

5.3 Conclusion

The rapid growth of digital payment systems has increased the need for intelligent fraud detection solutions capable of protecting users and financial institutions from cybercrime. Fraudulent activities continue to evolve, making traditional rule-based detection systems insufficient for modern payment environments.

The UPI Fraud Shield ML project successfully addresses this challenge by developing a machine learning-based fraud detection system capable of analyzing transaction behavior and identifying suspicious activities. Multiple machine learning algorithms were implemented and evaluated to determine their effectiveness in fraud detection.

The project successfully integrated:

- Data preprocessing techniques
- Feature engineering methods
- Machine learning models
- Fraud probability prediction
- SHAP explainability
- Analytics dashboards
- User authentication
- Prediction history management

The developed system provides an efficient and user-friendly solution for fraud prediction and analysis. The results obtained during testing demonstrate the effectiveness of machine learning techniques in identifying potentially fraudulent transactions.

The project also highlights the importance of explainable artificial intelligence in fraud detection applications. By providing transparent explanations for model predictions, users can better understand the factors contributing to fraud classification decisions.

Overall, the project achieved its objectives and demonstrated how machine learning can enhance the security of digital payment ecosystems.

5.4 Future Directions and Enhancements

Although the developed system performs effectively within its current scope, several enhancements can be implemented in future versions to improve functionality, scalability, and real-world applicability.

- **Real-Time Fraud Detection**

The current implementation operates on historical transaction datasets. Future versions can be integrated with real-time transaction streams to enable live fraud monitoring and instant detection.

- **Banking API Integration**

The system can be connected to banking APIs and payment gateways to automatically analyze transactions as they occur.

- **Cloud Deployment**

Deploying the application on cloud platforms such as AWS, Microsoft Azure, or Google Cloud Platform would improve accessibility, scalability, and performance.

- **Mobile Application Development**

A dedicated mobile application can be developed to provide fraud detection services on smartphones and tablets.

- **Deep Learning Integration**

Advanced deep learning architectures such as: Artificial Neural Networks

Long Short-Term Memory Networks (LSTM)

Transformer Models

can be incorporated to improve fraud detection performance.

- **Continuous Learning Mechanism**

Future systems can be designed to continuously retrain machine learning models using newly collected transaction data, enabling adaptation to evolving fraud patterns.

Multi-Language Support

- Support for multiple languages can improve accessibility and usability for a wider audience.

- **Advanced Security Features**

Additional security mechanisms may include:

Multi-Factor Authentication

Biometric Verification Behavioral

Authentication Device Fingerprinting

- **Fraud Alert System**

Future versions can automatically notify users through: Email alerts SMS notifications

Mobile push notifications when suspicious transactions are detected.

- **Integration with Financial Institutions**

The system can be extended for enterprise-level deployment within banks, financial institutions, and FinTech companies to support large-scale fraud monitoring operations.

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APPENDIX A

DATASET DESCRIPTION

A.1 Overview

The dataset used in this project is a synthetic fraud detection dataset designed to simulate real-world UPI and digital payment fraud scenarios. The dataset contains transaction-level, authentication-level, behavioral, and device-related information used to train machine learning models for fraud detection.

The objective of the dataset is to distinguish between legitimate and fraudulent transactions based on various transaction characteristics.

A.2 Dataset Statistics Parameter Value

Total Records	26,393
Total Features	47
Fraud Transactions	4,545
Legitimate Transactions	21,848
Target Variable	is_fraud

A.3 Target Variable

Value	Meaning
0	Legitimate Transaction
1	Fraudulent Transaction

A.4 Dataset Characteristics

The dataset contains:

- Transaction Information
- Authentication Information
- Device Information
- Behavioral Information
- UPI Handle Information
- User Interaction Information

The dataset was cleaned and preprocessed before model training.

APPENDIX B

FEATURE LIST

The following features were used in model training after preprocessing and feature selection.

S.No	Feature Name
1	amount
2	session_duration
3	authentication_attempts
4	receiver_account_age
5	receiver_transaction_history
6	transaction_amount_vs_sender_history
7	geographic_disparity
8	transaction_time_of_day
9	merchant_category_code
10	session_source
11	time_between_link_click_and_transaction
12	dns_lookup_age
13	recent_app_installs
14	input_timing_consistency
15	app_switching_frequency
16	keyboard_input_speed
17	input_pause_patterns

S.No	Feature Name
18	permissions_granted
19	screen_active_time
20	geographic_location_vs_ip
21	background_data_usage
22	recognized_screen_sharing_apps
23	time_between_otp_generation_and_input
24	pin_entry_speed
25	otp_request_frequency
26	otp_request_device_consistency
27	transaction_velocity
28	failed_transaction_count
29	authorization_method
30	transaction_type
31	request_description_keywords
32	request_amount_roundness
33	request_frequency
34	request_acceptance_rate
35	time_to_respond_to_request
36	requester_account_age
37	relationship_to_requester
38	upi_handle_age
39	handle_similarity_score
40	handle_contains_official_terms
41	handle_transaction_history
42	business_name_match

APPENDIX C

APPLICATION

1. Login Page
2. Registration Page
3. Prediction Dashboard
4. Fraud Prediction Result
5. Analytics Dashboard
6. ROC Curve
7. Feature Importance
8. SHAP Explainability
9. Prediction History

APPENDIX D

MODEL RESULTS

D.1 Performance Comparison

Model	Accuracy	Precision	Recall	F1 Score	AUC - ROC
Logistic Regression	0.9966	0.9912	0.9890	0.9901	0.9999
Decision Tree	0.9962	0.9795	0.9989	0.9891	0.9993
Random Forest	0.9996	0.9978	1.0000	0.9989	1.0000
Gradient Boosting	0.9998	0.9989	1.0000	0.9995	1.0000
Neural Network	0.9989	0.9945	0.9989	0.9967	1.0000

D.2 Best Performing Model

The Random Forest model demonstrated the best balance between prediction accuracy, robustness, feature importance analysis, and interpretability.

The model successfully identified fraudulent transactions and generated reliable fraud probability scores.

D.3 Important Features

Top features influencing fraud prediction:

1. Receiver Account Age
2. Receiver Transaction History
3. Transaction Amount

4. Request Amount Roundness
5. OTP Request Device Consistency
6. Input Timing Consistency
7. Business Name Match
8. Transaction Velocity
9. Request Frequency
10. Handle Similarity Score Insert
Feature Importance Graph.